

MACROSS: End-to-End Network Attack Detection Model for EV Charging Station Based on Temporal-Frequency Feature Fusion

Hailong Teng, Yan Tang, and Ziqiang Ma^(✉)

Faculty of Information Engineering, Ningxia University, Yinchuan 750000, China
tuki.t@foxmail.com, maziqiang@nxu.edu.cn

Abstract. As the global electric vehicle (EV) market grows, so does the frequency of cyberattacks on charging infrastructure, which lacks robust defense mechanisms. Existing network attack detection methods for EV charging stations, often based on machine learning, are limited by their reliance on manual feature engineering and are not optimized for handling large volumes of network traffic. Additionally, basic deep learning models that focus solely on temporal features are inadequate for identifying cyber threats. In this paper, we propose MACROSS, an innovative end-to-end detection model that directly processes raw network packets to extract both temporal and frequency domain features using convolutional neural networks (CNNs) and Fast Fourier Transform (FFT). An attention mechanism is employed to integrate these features, and a fusion feature extraction module based on residual networks learns the fused representation to perform a classification task that detects cyberattacks targeting EV charging stations. We conducted extensive experiments on the CICEVSE2024 dataset. The results demonstrate that MACROSS effectively identifies network attack traffic targeting EV charging stations and significantly outperforms baseline methods. Our model is publicly available for validation at <https://github.com/kisaragi-attention/macross>.

Keywords: EV Charging Station · End-to-End · Temporal-Frequency Feature Fusion · Neural Network · Traffic Detection · Attention.

1 Introduction

With the rapid growth of the global electric vehicle fleet and EV charging infrastructure, EV charging equipment is inevitably connected to the network for convenient management. Meanwhile, cyberattacks targeting EV charging equipment are also surging. As shown in Fig. 1, attacks targeting EV charging stations typically involve communication between EVs and charging stations via the ISO15118 protocol, and communication between charging stations and the Charging Station Management System (CSMS) via the Open Charge Point Protocol (OCPP). Attackers can achieve their goals of data theft, charging service disruption, and charging station hijacking by attacking either the EV charging

station or the CSMS. There have been multiple cases of cyberattacks targeting the connection between EVs and EV charging stations, leading to the theft of user credentials and disruption of EV charging. During the Russian invasion of Ukraine, EV charging stations along the M-11 highway between Moscow and St. Petersburg were remotely backdoored and their charging functions were controlled by a Ukrainian company, resulting in the disabling of charging stations along this route and the display of anti-Putin messages [1]. Researchers at the University of Oxford demonstrated that they could sniff EV data by using Software Defined Radio (SDR) to intercept communication traffic from EV charging stations [2]. The goal of EV charging station attack traffic detection is to identify attack traffic (e.g., DoS, Recon) and protect EV charging stations from data breaches, service disruption, and hijacking.

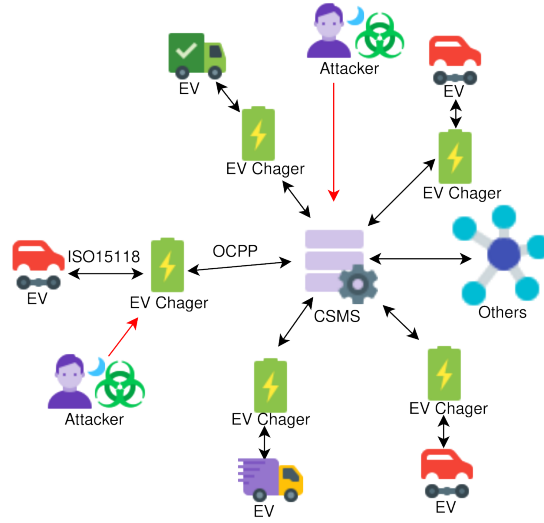


Fig. 1. Attacks on EV charging stations

Over the past decade, with the rapid advancement of artificial intelligence technology, various machine learning and deep learning algorithms have been widely applied in the field of cybersecurity. However, in-depth research on network attack detection for EV charging stations has not yet been conducted. Only a few studies have focused on this area, and these works generally use statistical features of EV charging station traffic to identify attack traffic. However, this approach requires manual feature engineering and an intermediate feature selection process, and exhibits limitations in addressing newly emerging types of cyberattacks. Therefore, this field currently faces the following challenges: (1) Many methods rely on statistical features, requiring manual extraction through feature engineering. With the rapid changes in the network environment, these statistical features may no longer be effective, leading to the loss of information

from the raw traffic. (2) Existing solutions all involve an intermediate feature selection process, which hinders efficient traffic processing. Furthermore, the quality of feature selection directly affects the detection performance of the models, lacking automatic feature extraction from the traffic.

In this paper, we propose an end-to-end EV charging stations attack traffic detection model to address the aforementioned challenges. This model directly processes raw traffic packets and leverages both temporal and frequency domain features of the traffic (packets) to extract valuable information, which is then used for a classification task to detect network attack traffic in EV charging stations.

The contributions of this paper can be summarized as follows:

- We introduce MACROSS, a novel end-to-end model for traffic detection in electric EV charging stations. By adopting an end-to-end approach, MACROSS automatically extracts rich information from raw traffic packets.
- We perform Fourier transformation on the raw traffic, concurrently extracting both temporal and frequency domain features. An attention mechanism is then employed to fuse these features, and a residual neural network is utilized to further learn the fused temporal-frequency features.
- We conducted experiments on the public dataset CICEVSE2024. The results demonstrate the effectiveness of our proposed method, achieving high accuracy and a low false positive rate.

The remainder of this paper is organized as follows: Section 2 reviews related work on traffic detection in EV charging stations; Section 3 details our proposed model; Section 4 presents the experiments and analyzes the results; and Section 5 concludes the paper.

2 Related Work

Currently, only a limited number of researchers have employed machine learning models and simple neural networks for attack traffic detection in EV charging stations. These methods commonly rely on statistical features for EV charging station traffic identification. Buedi et al. [3] were among the first to utilize machine learning algorithms for EV charging station attack traffic detection. They employed NFStream to extract features from raw traffic, using machine learning algorithms for binary and scenario classification of EV charging station attack traffic. They also collected and publicly released the CICEVSE2024 dataset. Hegde et al. [4] analyzed the frequency distribution of different attack types within EV charging station attack traffic, evaluated the importance of various statistical features for the model, and used classical and ensemble machine learning algorithms for EV charging station attack traffic detection. Janwiri [5] employed Principal Component Analysis (PCA) for feature selection in EV charging station traffic, evaluating the performance of machine learning models with 21, 15, 10, and 5 selected features, respectively, and identified the top 10 features with the greatest impact on EV charging station traffic classification.

Ullah et al. [6] utilized a deep swarm particle optimization (DSPO) algorithm for feature selection in EV charging station traffic and employed a multilayer perceptron (MLP) for binary and scenario classification of EV charging station traffic. Bamini et al. [7] used an enhanced dynamic whale threshold optimization (EDWPO) algorithm for feature selection in EV charging station traffic, evaluated feature importance, and employed machine learning models and simple neural networks for attack classification of EV charging station traffic.

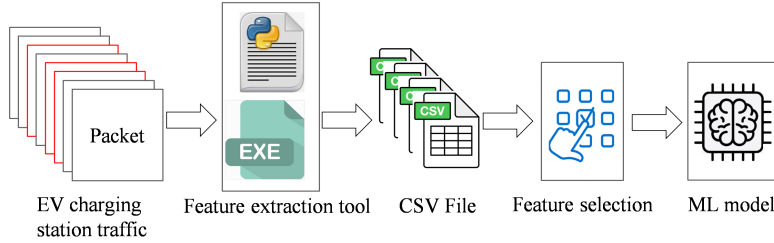


Fig. 2. Traditional EV charging station traffic detection process

The workflow of these methods is illustrated in Fig. 2. The first step involves extracting statistical features from raw traffic using Python scripts with traffic processing toolkits such as NFStream¹ or directly employing traffic feature extraction tools like CICFlowMeter². These features are then stored as CSV files. Subsequently, feature selection algorithms are applied to select relevant features, which are then fed into machine learning models or simple neural networks for EV charging station traffic identification. Alternatively, the extracted features can be directly input into machine learning models or simple neural networks for EV charging station traffic identification without undergoing feature selection. All these methods involve an intermediate feature selection process, and the quality of the model’s identification results depends on the manually selected features and the design of feature engineering, hindering efficient processing of raw EV charging station traffic. These methods commonly employ statistical features for EV charging station traffic identification, which can lead to information loss during the feature extraction process, preventing full exploitation of the information within the raw EV charging station traffic.

3 Methodology

We define a single packet as a sample, which is a common practice in many other types of traffic classification studies [8–11]. Our model aims to thoroughly and comprehensively mine the rich information within these packets. The overall

¹ <https://www.nfstream.org>

² <https://www.unb.ca/cic/research/applications.html>

model architecture is illustrated in Fig. 3 and mainly consists of packet pre-processing, a temporal feature extraction module, a frequency domain feature extraction module, a temporal-frequency feature fusion module, a fused feature extraction module, and a classification module.

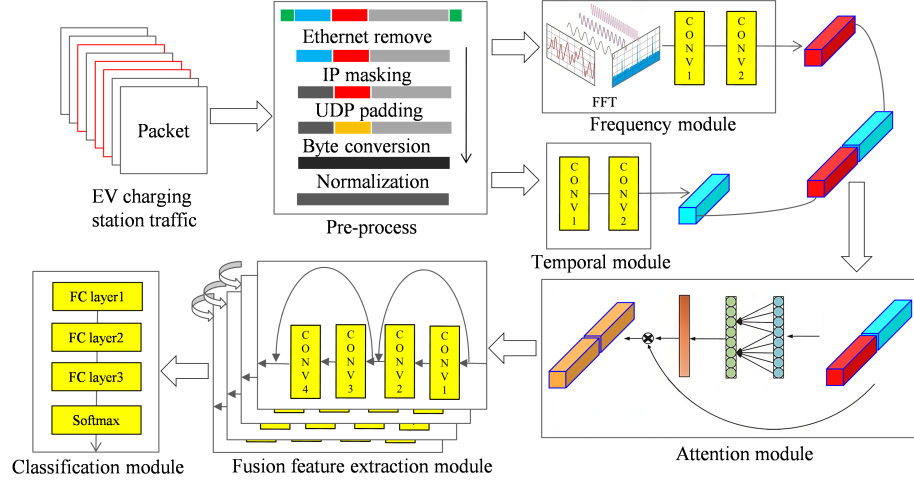


Fig. 3. The model architecture of MACROSS

3.1 Pre-process

We use raw packets as input to our model. The packets first undergo pre-processing, which mainly includes:

- 1) Ethernet remove (*Er*). Removing the Ethernet frame header and trailer from the packets. This is done because the Ethernet header does not provide useful information for the traffic identification task.
- 2) IP masking (*IPm*). Masking the IP addresses in the IP header. This prevents the neural network from potentially attempting to classify packets by learning IP addresses. Since the dataset is captured using a limited number of hosts and servers, masking IP addresses mitigates overfitting.
- 3) UDP padding (*UDPp*). Padding the UDP header to 20 bytes with zeros. TCP packets have a 20-byte header length, while UDP packets only have an 8-byte header length. To ensure consistent transport layer length, the UDP header is padded to match the TCP header length.
- 4) Byte conversion (*Bc*). Converting the raw packets into byte vectors. Since 1 byte equals 8 bits, values from *00000000* to *11111111* can be converted to numerical values between 0 and 255. Converting the eight-bit input to a single-digit input reduces the input size of the neural network and lowers the model's computational complexity.

- 5) Normalization (*Norm*). Normalizing the byte vectors by dividing each element by 255. This confines the values of each element in the byte vector to between 0 and 1. This normalization process helps accelerate model convergence and improve training stability.

The raw EV charging station traffic p , after preprocessing, yields a byte vector v_p .

$$v_p = \begin{cases} \text{Norm}(\text{Bc}(\text{UDPP}(\text{IPm}(\text{Er}(p)))))) & \text{if } \text{UDP packet} \\ \text{Norm}(\text{Bc}(\text{IPm}(\text{Er}(p)))) & \text{if } \text{TCP packet} \end{cases} \quad (1)$$

3.2 Frequency Module

To extract valuable features from the packets, we designed a frequency domain feature extraction module, allowing the model to learn the frequency domain characteristics of the packets. In this stage, the features within the packets are transformed into the frequency domain for learning. Inspired by this work [12], which mathematically proves that transforming traffic to the frequency domain results in only bounded information loss but can effectively improve the accuracy of identifying high-frequency attacks such as DoS, the frequency domain module employs two distinct convolutional kernels to extract frequency domain features from the packets. Each convolutional operation (*Conv*) includes a convolutional layer (all convolutional layers mentioned in this paper are one-dimensional convolutions) and a *ReLU* activation function, followed by a max pooling layer after the convolutional operations are completed.

The byte vector v_p is first transformed into the frequency domain through fast Fourier transform (*FFT*) and magnitude calculation (*Abs*), and then undergoes two convolutional operations and a max pooling layer to obtain the frequency domain vector f_p .

$$f_p = \text{Maxpool}(\text{Conv}(\text{Conv}(\text{Abs}(\text{FFT}(v_p))))) \quad (2)$$

3.3 Temporal Module

Temporal features of the packets also contain valuable information; therefore, we retained the temporal features of the packets. A temporal feature extraction module was designed to extract these features. The temporal module contains two distinct convolutional kernels to extract the temporal features of the packets. Each convolutional operation includes a convolutional layer and a *ReLU* activation function, followed by a max pooling layer after the convolutional operations are completed.

The byte vector v_p undergoes two convolutional operations and a max pooling layer to obtain the temporal vector t_p .

$$t_p = \text{Maxpool}(\text{Conv}(\text{Conv}(v_p))) \quad (3)$$

3.4 Attention Module

In recent years, channel attention mechanisms have demonstrated significant potential in improving the performance of convolutional networks. Attention mechanisms can assign different weight coefficients to different features, enabling neural networks to focus more on features that are beneficial for learning. Therefore, we employ the Efficient Channel Attention (*ECA*) mechanism [13] to fuse the temporal and frequency domain features, allowing the neural network to learn both the temporally and frequency domain features that are helpful for classification. The attention module takes the concatenated vector of the temporal and frequency modules as input. It first aggregates the convolutional features through global average pooling (*GAP*), then adaptively determines the kernel size k , performs a convolutional operation, and finally learns channel attention through a *Sigmoid* function.

There exists a mapping ϕ [13] between the channel dimension C and the kernel size k . Given the channel dimension C , we can determine the kernel size k through the inverse function ψ of ϕ .

$$C = \phi(k) = 2^{\gamma * k - b} \quad (4)$$

$$k = \psi(C) = \left\lfloor \frac{\log_2(C)}{\gamma} + \frac{b}{\gamma} \right\rfloor_{\text{odd}} \quad (5)$$

In Equation 6, x represents the input, which sequentially undergoes *GAP*, *Conv*, and *Sigmoid* operations to calculate the attention score. Finally, the calculated attention score is multiplied with the original input x through the *Multi* operation to complete the attention weighting.

$$ECA(x) = Multi(x, Sigmoid(Conv(GAP(x)))) \quad (6)$$

In Equation 7, the frequency domain vector f_p and the temporal vector t_p are concatenated through *Concat*, and then attention weighting is performed through *ECA* to obtain the fused vector a_{p1} .

$$a_{p1} = ECA(Concat(f_p, t_p)) \quad (7)$$

3.5 Fusion Feature Extraction Module

To further extract the fused temporal and frequency domain features, we decided to employ residual connections to learn these fused features. The fused feature extraction module is based on ResNet18 [14], with the key difference being that we removed the final classification layer of ResNet18, retaining only the feature extraction part for further extraction of the fused temporal and frequency domain features.

In Equation 8, the fused vector a_{p1} is processed by the residual module *Res* to obtain the fused feature extraction vector a_{p2} .

$$a_{p2} = Res(a_{p1}) \quad (8)$$

3.6 Classification Module

After obtaining the fused feature extraction vector a_{p2} , we input it into the classification module for multiple classification tasks. The classification module consists of a multilayer perceptron with three fully connected layers, followed by a *Softmax* operation to obtain the prediction result $\hat{y}$. Finally, we utilize the cross-entropy loss function to calculate the loss, which is then used for backpropagation during training.

$$\hat{y} = \text{Softmax}(\text{MLP}(a_{p2})) \quad (9)$$

$$\text{loss} = - \sum_{i=1}^n y_i \log \hat{y}_i \quad (10)$$

4 Experiment and Analysis

In this section, we present our experimental setup, including the dataset, implementation details, and evaluation metrics. We then conduct a series of experiments, including comparative experiments and ablation studies, to thoroughly and comprehensively evaluate the performance of our model.

4.1 Experiment Setup

Dataset. We utilize the public dataset CICEVSE2024, which contains network attack traffic, hardware event logs during attacks, and energy consumption records collected from real EV charging stations. We focus specifically on the network attack traffic portion, which includes both benign and attack traffic under charging and idle states of the EV charging stations. Specific details are shown in Table 1.

Implementation Details and Evaluation Metrics. In the preprocessing stage, we limit the length of the byte vectors to 1500, truncating byte vectors exceeding this length and padding shorter byte vectors to 1500. This is due to the Maximum Transmission Unit (MTU) limitation of 1500 bytes in computer networks. In the temporal and frequency domain feature extraction modules, the size of the first convolutional kernel is set to 4 with a stride of 3 and a quantity of 50, while the size of the second convolutional kernel is set to 5 with a stride of 1 and a quantity of 50. In the attention module, $\gamma = 2$ and $b = 1$. In the fused feature extraction module, the size of each convolutional kernel is 3 with a stride of 1. We have open-sourced the model code on GitHub, and more details can be found at <https://github.com/kisaragi-attention/macross>. All implementations are based on Python 3.10, PyTorch 2.4.1, and pytorch-lightning 1.7.7, using an NVIDIA RTX 3090 GPU.

In classification tasks, commonly used evaluation metrics are Accuracy, Precision, Recall, and F1-score. For fair comparison, we use these four metrics to evaluate our model and the baselines.

Table 1. CICEVSE2024 dataset label

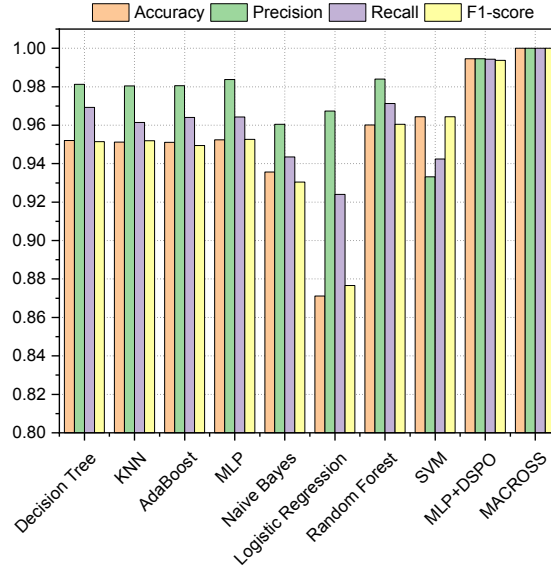
Binary label	Scenario label	Attack label
Benign	Benign	Benign
Attack	Recon	TCP Port Scan
		Service Version Detection
		OS Fingerprinting
		Aggressive Scan
		SYN Stealth Scan
		Vulnerability Scan
		Slowloris Scan
	DoS	UDP Flood
		ICMP Flood
		PSHACK Flood
		ICMP Fragmentation
		TCP Flood
		SYN Flood
		SynonymousIP Flood

4.2 Comparison Experiments

We designed three classification tasks to validate our model: binary classification, scenario classification, and attack classification. For the binary and scenario classification tasks, we used classical machine learning models as baselines, including Decision Tree, K-Nearest Neighbors (KNN), Adaptive Boosting (AdaBoost), MLP, Naive Bayes, Logistic Regression, Random Forest, and Support Vector Machine (SVM). These baseline models all employed a traditional detection approach, extracting statistical features from raw packets using feature extraction tools, storing them in CSV files, and then inputting them into the models for classification. In addition to comparing with the aforementioned baseline models, we also compared our model with the work presented in [6], which uses a deep particle swarm optimization algorithm for feature selection. The comparison results of the binary and scenario classification experiments are shown in Fig. 4 and Table 2. Our model achieved the best performance. Specifically, in the binary classification task, our model achieved 100% performance on all four metrics. In the scenario classification task, our model also outperformed the baseline methods and achieved the best performance.

For the attack classification task, we used SVM, Random Forest, MLP, and CNN without feature selection, and SVM, Random Forest, MLP, and CNN with feature selection using an enhanced dynamic threshold whale optimization algorithm as our baseline models. The experimental comparison results are shown in Table 3. Our model also achieved the best performance in this task.

Based on the results above, we can draw the following conclusions: (1) Our model achieved the best performance compared to several baselines across all four metrics. This comprehensive performance demonstrates the effectiveness of our proposed method. (2) The performance of traditional machine learning models is inferior to our model. This is because these methods rely on statistical

**Fig. 4.** Comparison results on binary classification tasks**Table 2.** Comparison results on scenario classification tasks

Model	Accuracy	Precision	Recall	F1-score
Decision Tree	93.73%	96.93%	93.73%	94.77%
KNN	93.68%	96.88%	93.68%	94.72%
AdaBoost	93.76%	96.97%	93.76%	94.80%
MLP	90.56%	93.23%	90.56%	91.47%
Naive Bayes	92.13%	95.01%	92.13%	93.09%
Logistic Regression	82.61%	94.29%	82.61%	87.27%
Random Forest	93.73%	96.94%	93.73%	94.77%
SVM	94.12%	91.50%	94.12%	92.79%
MLP+DSPO	99.29%	99.28%	99.24%	98.98%
MACROSS	99.84%	99.84%	99.84%	99.84%

Table 3. Comparison results on attack classification tasks

Model	Accuracy	Precision	Recall	F1-score
SVM	79.71%	72.68%	79.71%	74.62%
Random Forest	88.15%	87.75%	88.15%	87.85%
MLP	83.50%	81.64%	83.50%	81.89%
CNN	82.22%	81.93%	82.28%	80.46%
SVM+EDTWO	89.19%	90.30%	89.19%	89.07%
Random Forest+EDTWO	92.41%	92.63%	92.41%	92.47%
MLP+EDTWO	89.17%	90.06%	89.17%	88.84%
CNN+EDTWO	86.85%	88.07%	86.85%	86.24%
MACROSS	98.40%	98.40%	98.40%	98.40%

features, and simple deep learning models that rely solely on temporal features also perform worse than our model. This highlights the effectiveness of combining temporal and frequency domain features to capture the characteristics of EV charging station traffic. (3) Although advanced feature selection methods, such as deep particle swarm optimization and enhanced dynamic threshold whale optimization, are employed in the feature selection process of traditional detection approaches, the models still fail to achieve ideal results. The main reason for this is that traditional detection approaches involve an intermediate feature selection process, which hinders efficient processing of EV charging station traffic.

4.3 Ablation Experiments

To examine the effectiveness of each component in MACROSS, we conducted ablation studies on the following aspects: (1) MACROSS-FF: To demonstrate the effectiveness of frequency domain features (FF), only temporal feature extraction was performed. (2) MACROSS-TF: To demonstrate the effectiveness of temporal features (TF), only frequency domain feature extraction was performed. (3) MACROSS-Attention: To demonstrate the effectiveness of feature fusion, the attention-based feature fusion module was removed, and features were simply concatenated. (4) MACROSS-FFE: To demonstrate the effectiveness of fused feature extraction (FFE), the fused feature extraction module was removed.

The results of the ablation studies are shown in Table 4. We can observe that removing the frequency domain features has the most significant negative impact on the model’s performance. This demonstrates the importance of frequency domain features for EV charging station traffic identification. Removing the temporal features resulted in a 7.2% decrease in the model’s accuracy, indicating that temporal features are also helpful for EV charging station traffic identification. Removing the attention module led to a 2.2% decrease in the model’s accuracy, demonstrating that the attention mechanism can facilitate effective fusion of temporal and frequency domain features. Removing the fused feature extraction module resulted in a 6.3% decrease in the model’s accuracy, demonstrating that further extraction of the fused temporal and frequency domain features can effectively improve the accuracy of EV charging station traffic identification.

Table 4. Ablation results of key components in MACROSS

Model	Accuracy	Precision	Recall	F1-score
MACROSS-FF	79.4969%	74.4980%	79.4968%	75.4997%
MACROSS-TF	91.1950%	90.5640%	91.1949%	89.1776%
MACROSS-Attention	96.1524%	96.6247%	96.1524%	96.0621%
MACROSS-FFE	92.0755%	91.9019%	92.0754%	91.9791%
MACROSS	98.4018%	98.4040%	98.4017%	98.4014%

5 Conclusion

In this paper, we propose MACROSS, an end-to-end EV charging station traffic detection model based on temporal-frequency feature fusion. We directly extract meaningful features from raw EV charging station traffic through an end-to-end approach, eliminating the intermediate feature selection process and improving the automation and detection performance of existing detection methods. Specifically, MACROSS encodes raw EV charging station traffic packets into byte vectors, utilizes Fourier transform and convolutional operations to extract temporal and frequency domain features from the byte vectors, and employs an attention mechanism to fuse the temporal and frequency domain features. Residual connections are used to further learn the fused temporal-frequency features. A series of experiments demonstrate the effectiveness of our proposed method.

Acknowledgement

This work is supported by the General Project of Key R&D Program in Ningxia Hui Autonomous Region of China under grant 2022BDE03008.

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